2VGI | pdb_00002vgi

HUMAN ERYTHROCYTE PYRUVATE KINASE: R486W MUTANT

PDB DOI: https://doi.org/10.2210/pdb2VGI/pdb
Entry: 2VGI **supersedes:** 1LIX

Classification: TRANSFERASE
Organism(s): Homo sapiens
Expression System: Escherichia coli BL21
Mutation(s): Yes

Deposited: 2007-11-13 **Released:** 2007-11-20
Deposition Author(s): Valentini, G., Chiarelli, L., Fortin, R., Dolzan, M., Galizzi, A., Abraham, D.J., Wang, C., Bianchi, P., Zanella, A., Mattevi, A.

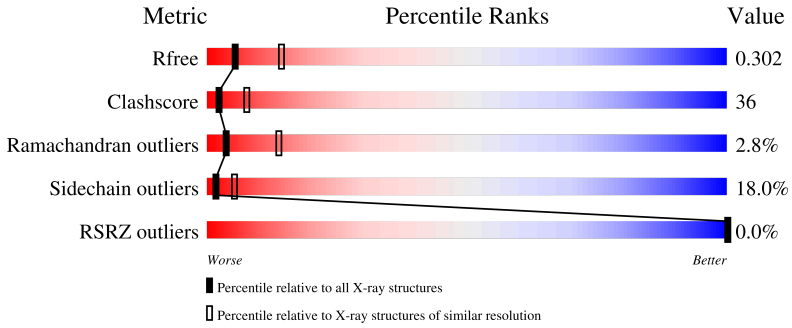
Experimental Data Snapshot

Method: X-RAY DIFFRACTION
Resolution: 2.87 Å
R-Value Free:
0.311 (Depositor), 0.302 (DCC)
R-Value Work:
0.257 (Depositor)
R-Value Observed:
0.258 (Depositor)

Starting Model: experimental
View more details

wwPDB Validation

3D Report | Full Report



Ligand Structure Quality Assessment



This is version 2.0 of the entry. See complete history.

Literature

Structure and Function of Human Erythrocyte Pyruvate Kinase. Molecular Basis of Nonspherocytic Hemolytic Anemia.

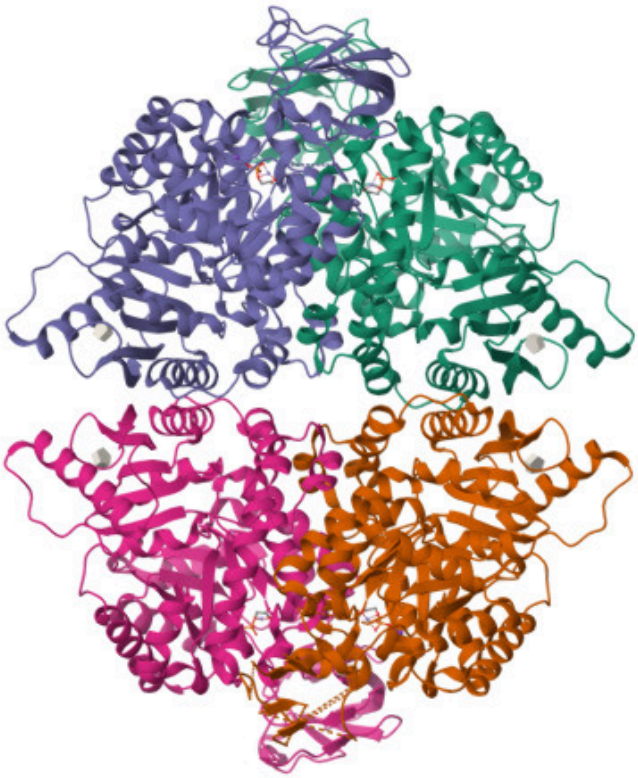
Valentini, G., Chiarelli, L.R., Fortin, R., Dolzan, M., Galizzi, A., Abraham, D.J., Wang, C., Bianchi, P., Zanella, A., Mattevi, A.
(2002) J Biol Chem **277**: 23807

PubMed: 11960989
DOI: https://doi.org/10.1074/jbc.M202107200
Primary Citation Related Structures:
2VGB, 2VGF, 2VGG, 2VGI

PubMed Abstract:
Deficiency of human erythrocyte isozyme (RPK) is, together with glucose-6-phosphate dehydrogenase deficiency, the most common cause of the nonspherocytic hemolytic anemia. To provide a molecular framework to the disease, we have solved the 2.7 A resolution crystal structure of human RPK in complex...
View More

Organizational Affiliation:
Dipartimento di Genetica e Microbiologia, Università di Pavia, via Abbiategrasso 207, 27100 Pavia, Italy. giovale@unipv.it

Biological Assembly 1



Explore in 3D: Structure | Sequence Annotations | Electron Density | Validation Report | Ligand Interaction (FBP)

Global Symmetry: Dihedral - D2 (Explore in 3D)
Global Stoichiometry: Homo 4-mer - A4

Find Similar Assemblies

Biological assembly 1 assigned by authors and generated by PISA (software)

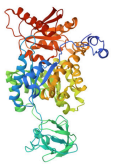
Macromolecule Content

- Total Structure Weight: 230.37 kDa
- Atom Count: 15,342
- Modeled Residue Count: 2,008
- Deposited Residue Count: 2,112
- Unique protein chains: 1

Macromolecules

Find similar proteins by: Sequence | 3D Structure

Entity ID: 1

Molecule	Chains	Sequence Length	Organism	Details	Image
PYRUVATE KINASE ISOZYMES R/L	A, B, C, D	528	Homo sapiens	**Mutation(s)**: 1 **EC:** 2.7.1.40	

UniProt & NIH Common Fund Data Resources

Find proteins for P30613 *(Homo sapiens)* Explore P30613 Go to UniProtKB: P30613

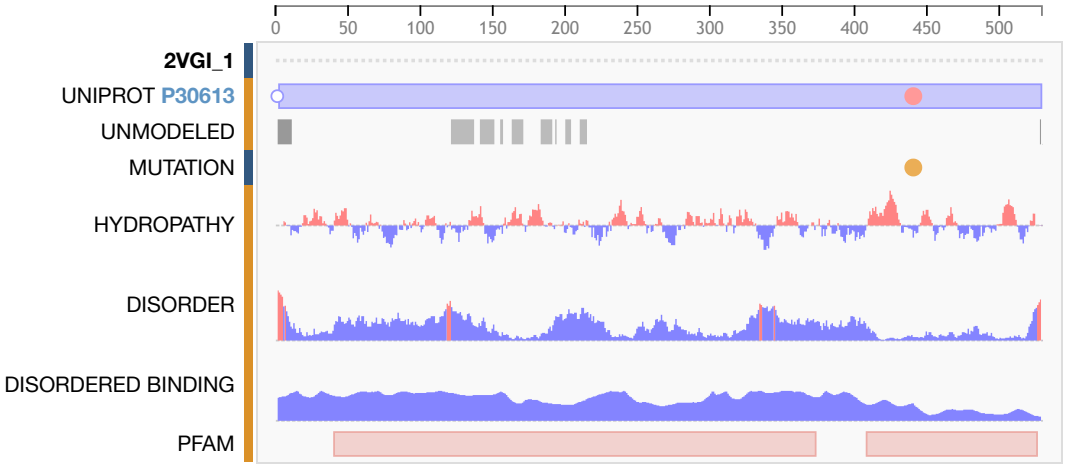
PHAROS: P30613 GTEx: ENSG00000143627

Entity Groups

Sequence Clusters	30% Identity, 50% Identity, 70% Identity, 90% Identity, 95% Identity, 100% Identity
UniProt Group	P30613

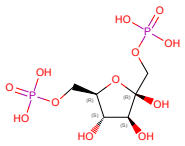
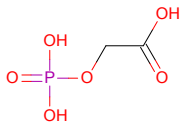
Sequence Annotations Expand

Reference Sequence 2VGI_1



Small Molecules

Ligands 4 Unique

ID	Chains	Name / Formula / InChI Key	2D Diagram	3D Interactions
FBP Download: Instance Coordinates	E [auth A], I [auth B], M [auth C], Q [auth D]	**1,6-di-O-phosphono-beta-D-fructofuranose** $C_6 H_{14} O_{12} P_2$ RNBGYGVWRKECFJ-ARQDHWQXSA-N		Interactions Interactions & Density
PGA Download: Instance Coordinates	F [auth A], J [auth B], N [auth C], R [auth D]	**2-PHOSPHOGLYCOLIC ACID** $C_2 H_5 O_6 P$ ASCFNMCAHFUBCO-UHFFFAOYSA-N		Interactions Interactions & Density
MN Download: Instance Coordinates	H [auth A], L [auth B], P [auth C], T [auth D]	**MANGANESE (II) ION** Mn WAEMQWOKJMHJLA-UHFFFAOYSA-N	Mn^{+2}	Interactions Interactions & Density
K Download: Instance Coordinates	G [auth A], K [auth B], O [auth C], S [auth D]	**POTASSIUM ION** K NPYPAHLBTDXSSS-UHFFFAOYSA-N	K^{+}	Interactions Interactions & Density

Experimental Data & Validation

Experimental Data

Method: X-RAY DIFFRACTION
Resolution: 2.87 Å
R-Value Free: 0.311 (Depositor), 0.302 (DCC)
R-Value Work: 0.257 (Depositor)
R-Value Observed: 0.258 (Depositor)

Space Group: P 1 2_1 1

Unit Cell:

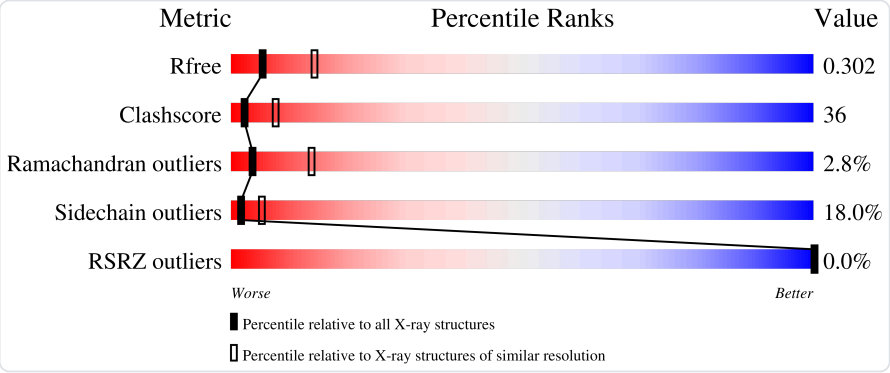
Length (Å)	Angle (°)
a = 73.705	α = 90
b = 171.164	β = 91.61
c = 85.05	γ = 90

Software Package:

Software Name	Purpose
REFMAC	refinement
MOSFLM	data reduction
CCP4	data scaling
CCP4	phasing

Structure Validation

View Full Validation Report



Ligand Structure Quality Assessment



Entry History

Deposition Data

Released Date: 2007-11-20
Deposition Author(s): Valentini, G., Chiarelli, L., Fortin, R., Dolzan, M., Galizzi, A., Abraham, D.J., Wang, C., Bianchi, P., Zanella, A., Mattevi, A.
This entry supersedes: 1LIX

Revision History (Full details and data files)

- **Version 1.0: 2007-11-20**
  Type: Initial release
- **Version 1.1: 2011-07-13**
  Changes: Refinement description, Version format compliance
- **Version 1.2: 2019-11-20**
  Changes: Advisory, Derived calculations, Other
- **Version 1.3: 2020-07-29**
  Type: Remediation
  Reason: Carbohydrate remediation
  Changes: Data collection, Derived calculations, Structure summary
- **Version 1.4: 2023-12-13**
  Changes: Data collection, Database references, Refinement description, Structure summary
- **Version 2.0: 2024-05-01**
  Changes: Atomic model, Derived calculations

Follow us:

Partnerships & Collaborators:

PDB-101 · WORLDWIDE PDB PROTEIN DATA BANK · EMDataResource · NAKB · wwPDB Foundation · PDB-IHM

RCSB PDB Core Operations are funded by the U.S. National Science Foundation (DBI-2321666), the US Department of Energy (DE-SC0019749), and the National Cancer Institute, National Institute of Allergy and Infectious Diseases, and National Institute of General Medical Sciences of the National Institutes of Health under grant R01GM157729. RCSB PDB uses resources of the National Energy Research Scientific Computing Center (NERSC), a Department of Energy User Facility.